

host_agent

Show me the houses in Exeter which is likely hard to heat

Tool Calls

send_message

- target: planning_agent
- task_description: Show me the houses in Exeter which are likely hard to heat. Provide the general steps to solve this geospatial query.

planning_agent

The area is Exeter, so there will be 1 entry for Exeter. The entities to search for are houses in Exeter. The condition is to identify houses that are likely hard to heat, which may require specific criteria or data related to heating efficiency or insulation.

output steps: ["Find Exeter 1 area", "Search for houses in Exeter as many search results", "Apply conditions to identify houses likely hard to heat based on heating efficiency or insulation criteria"]

Tool Calls

send_message

- target: named_area
- task_description: Find the area polygon for Exeter.

named_area

I found the area polygon for Exeter. The filtered results are saved in the artifact named `filtered_exeter_area_polygon`. This GeoDataFrame contains only the polygon(s) representing Exeter.

Tool Calls

call_os_ngd

- filters: None
- bbox: None
- polygon_or_point: True
- street_address: None
- filename: exeter_area_polygon

send_message

- target: data_analysis_agent
- task_description: Filter the artifact 'exeter_area_polygon' to find the specific polygon for Exeter. Save the filtered results as 'filtered_exeter_area_polygon'.

data_analysis_agent

Filtered to 34 polygon(s) for Exeter. - Artifact name: filtered_exeter_area_polygon - Description: GeoDataFrame containing only the polygon(s) representing Exeter, filtered using name and description columns.

Tool Calls

generate_metadata_for_artifacts

- artifact_names: ['exeter_area_polygon']

Tool Code Executor

```
def filter_exeter_area_polygon(data: list):
    gdf = data[0]
    # Filter for Exeter using multiple relevant columns
    filtered = gdf[((gdf['name1_text'].str.lower() == 'exeter') |
                    (gdf['description'].str.lower().str.contains('city')) |
                    (gdf['descriptiongroup'].str.lower().str.contains('settlement')) |
                    (gdf['name1_text'].str.lower().str.contains('exeter')))]
    return [
        f"Filtered to {len(filtered)} polygon(s) for Exeter.",
```

```
        'filtered_exeter_area_polygon',
        'GeoDataFrame containing only the polygon(s) representing Exeter, filtered using name and description columns.',
        filtered
    ]
```

named_area

I found the area polygon for Exeter. The filtered results are saved in the artifact named `filtered_exeter_area_polygon`. This GeoDataFrame contains only the polygon(s) representing Exeter.

Tool Calls

send_message

- target: buildings
- task_description: Search for houses in the area defined by the artifact 'filtered_exeter_area_polygon'. Focus on residential buildings or houses.

buildings

Search Results: - 40,890 houses have been identified in the area defined by the artifact 'filtered_exeter_area_polygon'. - These houses are filtered from the total 72,895 buildings in the area.

Definition Used: - A house is defined as a building where `buildinguse_addresscount_residential > 0` and `buildinguse_addresscount_total = 1`.

Artifact Generated: - `houses_in_filtered_exeter_area`: This artifact contains all buildings in the filtered Exeter area that are classified as houses, with all original columns retained.

If you need further analysis or details about these houses, let me know!

Tool Calls

call_os_ngd

- filters: None
- bbox: `filtered_exeter_area_polygon`

- polygon_or_point: True
- street_address: None
- filename: all_buildings_in_filtered_exeter_area

send_message

- target: data_analysis_agent
- task_description: From the artifact 'buildings_all_buildings_in_filtered_exeter_area', filter and extract all houses. A house is defined as a building where buildinguse_addresscount_residential > 0 and buildinguse_addresscount_total = 1. Save the filtered results as 'houses_in_filtered_exeter_area'.

data_analysis_agent

Filtered houses in Exeter area: 40,890 entries out of 72,895 total buildings. An artifact has been generated with the name 'houses_in_filtered_exeter_area'. This GeoDataFrame contains all buildings in the filtered Exeter area that are classified as houses, defined as buildings where 'buildinguse_addresscount_residential' > 0 and 'buildinguse_addresscount_total' = 1. All original columns are retained.

Tool Calls

generate_metadata_for_artifacts

- artifact_names: ['buildings_all_buildings_in_filtered_exeter_area']

Tool Code Executor

```
def extract_houses_in_filtered_exeter_area(data: list):  
    # Extract the first (and only) GeoDataFrame from the list  
    gdf = data[0]  
    # Filter for houses: residential > 0 and total = 1  
    houses_gdf = gdf[(gdf['buildinguse_addresscount_residential'] > 0) & (gdf['buildinguse_addresscount_total'] == 1)]  
    summary = f"Filtered houses in Exeter area: {len(houses_gdf)} entries out of {len(gdf)} total buildings."  
    artifact_name = 'houses_in_filtered_exeter_area'  
    artifact_description = (  
        "GeoDataFrame containing all buildings in the filtered Exeter area that are classified as houses. "  
        "A house is defined as a building where 'buildinguse_addresscount_residential' > 0 and "  
        "'buildinguse_addresscount_total' = 1. All original columns are retained."  
    )  
    return [summary, artifact_name, artifact_description, houses_gdf]
```

buildings

Search Results: - 40,890 houses have been identified in the area defined by the artifact 'filtered_exeter_area_polygon'. - These houses are filtered from the total 72,895 buildings in the area.

Definition Used: - A house is defined as a building where buildinguse_addresscount_residential > 0 and buildinguse_addresscount_total = 1.

Artifact Generated: - houses_in_filtered_exeter_area: This artifact contains all buildings in the filtered Exeter area that are classified as houses, with all original columns retained.

If you need further analysis or details about these houses, let me know!

Tool Calls

generate_metadata_for_all_artifacts

- args: {}

send_message

- target: buildings
- task_description: From the artifact 'houses_in_filtered_exeter_area', identify houses that are likely hard to heat. Use available building attributes such as age, construction type, insulation, or any other relevant features that may indicate poor heating efficiency.

buildings

From the artifact houses_in_filtered_exeter_area, 35,348 houses have been identified as likely hard to heat out of a total of 40,890 houses in the Exeter area.

Criteria Used: - Houses built pre-1999 (older buildings are more likely to be hard to heat). - Houses with construction materials such as timber, non-standard, mixed, unknown, or mobile home types, which are typically less energy efficient.

Artifact Generated: - hard_to_heat_houses_in_exeter_area: This artifact contains all houses in the Exeter area that are likely to be hard to heat, based on the criteria above. All original columns are retained.

If you need further analysis or a summary of these hard-to-heat houses, let me know!

Tool Calls

send_message

- target: data_analysis_agent
- task_description: From the artifact 'houses_in_filtered_exeter_area', identify houses that are likely hard to heat. Use available building attributes such as age (older buildings, especially pre-1999, are more likely to be hard to heat), construction type, insulation, or any other relevant features that may indicate poor heating efficiency. Save the filtered results as 'hard_to_heat_houses_in_exeter_area'.

data_analysis_agent

Identified 35,348 likely hard to heat houses out of 40,890 total houses in the Exeter area. An artifact has been generated with the name 'hard_to_heat_houses_in_exeter_area'. This GeoDataFrame contains all houses in the Exeter area that are likely to be hard to heat, based on age (pre-1999) or construction material (timber, non-standard, mixed, unknown, or mobile home). All original columns are retained.

Tool Calls

generate_metadata_for_artifacts

- artifact_names: ['houses_in_filtered_exeter_area']

Tool Code Executor

```
def identify_hard_to_heat_houses(data):  
    import pandas as pd  
    import geopandas as gpd  
    gdf = data[0]  
    pre_1999_periods = [  
        'Pre-1919', '1919-1944', '1870-1918', '1960-1979', '1945-1959', '1980-1989', '1990-1999', '1837-1869', 'Pre-1837'  
    ]  
    poor_construction_types = [  
        'Timber Or Wood', 'Other Artificial Material (Not Concrete)', 'Other Non-Standard Or System Build',  
        'Mixed (Masonry And Timber)', 'Mixed (Masonry And Metal)', 'Mixed (Plaster And Timber)', 'Unknown',  
        'Static Caravan Or Mobile Home'  
    ]
```



```
mask_age = gdf['buildingage_period'].isin(pre_1999_periods)
mask_material = gdf['constructionmaterial'].isin(poor_construction_types)
hard_to_heat = gdf[mask_age | mask_material].copy()
summary = f"Identified {len(hard_to_heat)} likely hard to heat houses out of {len(gdf)} total houses in Exeter area."
artifact_name = 'hard_to_heat_houses_in_exeter_area'
artifact_description = (
    "GeoDataFrame of houses in Exeter area likely to be hard to heat, based on age (pre-1999) "
    "or construction material (timber, non-standard, mixed, unknown, or mobile home). All original columns retained."
)
return [summary, artifact_name, artifact_description, hard_to_heat]
```

plotting_agent

Plotted 35,348 houses likely hard to heat in Exeter. First 5 entries:

	description	buildingage_period	constructionmaterial	geometry_area_m2
45743	Detached House	Pre-1919	Brick Or Block Or Stone	219.717
25225	Detached House	Pre-1919	Brick Or Block Or Stone	134.063
67277	Detached House	Pre-1919	Brick Or Block Or Stone	248.278
61266	Semi-Detached House	1919-1944	Brick Or Block Or Stone	48.673
60956	Semi-Detached House	1919-1944	Brick Or Block Or Stone	50.711

An interactive folium map has been generated showing the distribution of houses in Exeter likely hard to heat, colored orange.

Artifact name: hard_to_heat_houses_exeter_map

Artifact description: Folium map showing the distribution of houses in Exeter likely hard to heat, colored orange.

Artifact file: hard_to_heat_houses_exeter_map.html

Tool Calls

generate_metadata_for_artifacts

- artifact_names: ['hard_to_heat_houses_in_exeter_area']

Tool Code Executor

```
def plot_hard_to_heat_houses_distribution(data: list):
    import folium
    import geopandas as gpd
```

```

import pandas as pd
from shapely.geometry import mapping

gdf = data[0].copy()
# Remove columns with non-serializable types (like Timestamp)
for col in gdf.columns:
    if pd.api.types.is_datetime64_any_dtype(gdf[col]):
        gdf[col] = gdf[col].astype(str)
# Ensure CRS is EPSG:4326 for folium
if gdf.crs is not None and gdf.crs.to_epsg() != 4326:
    gdf = gdf.to_crs(epsg=4326)
elif gdf.crs is None:
    gdf.set_crs(epsg=4326, inplace=True)

# Center map on Exeter (approximate center from data)
centroid = gdf.geometry.centroid.unary_union.centroid
m = folium.Map(location=[centroid.y, centroid.x], zoom_start=13, tiles='cartodbpositron')

# Add polygons to map
folium.GeoJson(
    gdf,
    name='Hard to Heat Houses',
    style_function=lambda x: {
        'fillColor': '#ff7800',
        'color': '#ff7800',
        'weight': 0.5,
        'fillOpacity': 0.4
    },
    tooltip=folium.GeoJsonTooltip(fields=['description', 'buildingage_period', 'constructionmaterial', 'geometry_area_m2'],
                                  aliases=['Type', 'Age Period', 'Material', 'Area (m2)'])
).add_to(m)

# Save map
map_filename = 'hard_to_heat_houses_exeter_map.html'
m.save(map_filename)

# Prepare summary
summary = f"Plotted {len(gdf)} houses likely hard to heat in Exeter. First 5 entries:\n" + \
    str(gdf[['description', 'buildingage_period', 'constructionmaterial', 'geometry_area_m2']].head(5))
artifact_name = 'hard_to_heat_houses_exeter_map'
artifact_description = 'Folium map showing the distribution of houses in Exeter likely hard to heat, colored orange.'
return [summary, artifact_name, artifact_description, map_filename]

```

host_agent

I have identified and mapped the houses in Exeter that are likely hard to heat. The criteria used include houses built before 1999 and those constructed with materials such as timber, non-standard, mixed, unknown, or mobile home types, which are typically less energy efficient.

- 35,348 houses in Exeter are classified as likely hard to heat.
- These are distributed across the city and are visualized on an interactive map.

Would you like to view the interactive map or need a summary of specific areas within Exeter? Let me know if you want further details or analysis!